

Implementing Gradient Descent

Gradient Descent Algorithm:

On each step w is updated to:-

$$w = w - \alpha \frac{d}{dw} J(w, b)$$

what this expression is saying,

- "update your parameter w by taking current value of w and adjusting it a small amount, which is this expression on the right $\left[-\alpha \frac{d}{dw} J(w, b) \right]$ "

Breakdown:

$=$: assignment operator

(alpha) α : learning rate (a small +ve number between 0 & 1)
it controls how big of a step you take downhill.

$\frac{d}{dw} J(w, b)$: Derivative term of cost function J
↳ for now it tells in which direction you want to take your baby step in.

Remember, there's also b . You also have assignment operations on b .

$$b = b - \alpha \frac{d}{db} J(w, b)$$

→ For gradient descent algorithm, you are going to repeat these two update steps until algorithm converges.

↳ By converges, It means that you reach point at local minimum where the parameters w & b no longer change much with each additional step you take.

For G.D a subtle detail:

"You are gonna update both params simultaneously (at the same time).

Correct Simultaneous Update:

$$1) \cdot \text{temp_}w = w - \alpha \frac{d}{dw} J(w, b)$$

$$2) \cdot \text{temp_}b = b - \alpha \frac{d}{db} J(w, b)$$

$$3) \cdot w = \text{temp_}w$$

$$4) \cdot b = \text{temp_}b$$

Correct seq

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

$2 \rightarrow 1 \rightarrow 3 \rightarrow 4$

Incorrect seq

$1 \rightarrow 3 \rightarrow 2 \rightarrow 4$

$2 \rightarrow 4 \rightarrow 1 \rightarrow 3$